

Bachelor in Computer Applications

Course Title: **Database Management System**

Code No: Semester: **3**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

1. What is Transaction? Draw a Transaction state diagram and discuss the typical state that a transaction goes through during execution.
2. What is a functional dependency? What are the four informal measures of quality for relational schema design? Explain.
3. Explain the various Symbols used in E-R Model?
4. What is NoSQL? Explain the NOSQL characteristics related to data models and query languages.
5. What is the system log used for? What are the typical kinds of entries in a system log?
6. What is normalization? Explain 2NF with example.
7. What is meant by the concurrent execution of database transactions in a multiuser system? Discuss why concurrency control is needed, and give informal examples.
8. Retrieve the Employee Name, Department Number and Department Name of "John Smith" using Relational Algebra. The relations are given below.
a. EMPLOYEE (Ssn, Fname, Lname, Bdate, Address, Sex, Salary, SuperSSN, Dno)
b. PROJECT (Pnumber, Pname, Plocation, Dnum)
c. WORKS ON (Essn, Pno, Hours)
d. DEPARTMENT (Dname, Dnumber, Mgr_ssn, Mgr Start date)
9. Discuss the different types of failures. What is meant by catastrophic failure?
10. What is Relational Data model? How Relation is different than Flat file?
11. What is Database? Explain the characteristics of Database Approach.
12. What are the three layers of the Three-Schema Architecture? Explain.